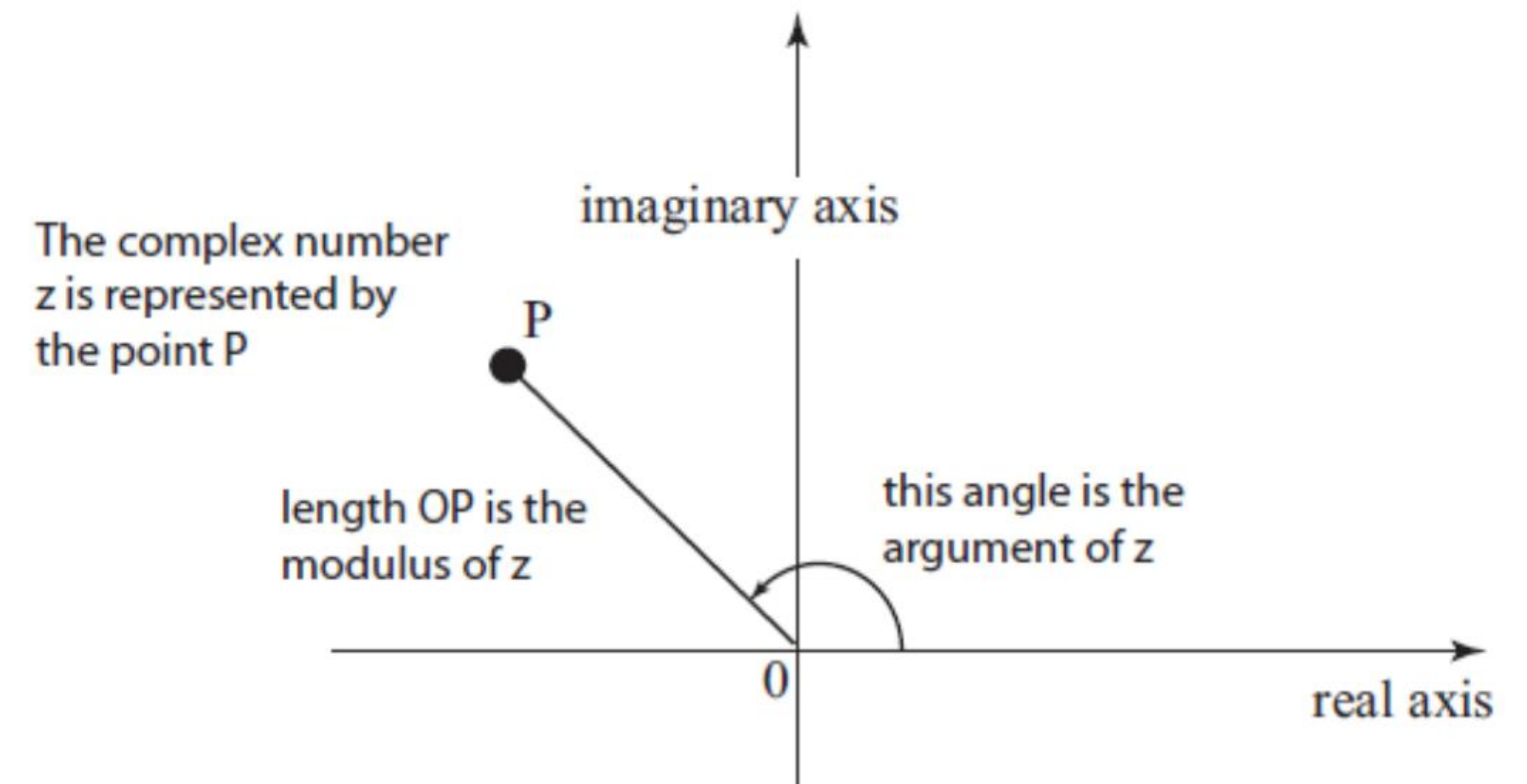


Geometry and Locus of Complex Numbers

A Step-by-Step Problem Solving Masterclass | Chapter 2, Section 2.6

PRESENTATION OVERVIEW

- **1. Core Concept:** The Circle in the Argand Plane
- **2. Solved Examples:**
 - 2.18: Triangles in the Complex Plane
 - 2.19 & 2.20: Identifying Circles & Interiors
 - 2.21: Finding the Cartesian Locus
- **3. Exercise 2.6 Solutions:**
 - Q1 & Q2: Ratios and Imaginary Parts
 - Q3: Extracting Real & Imaginary Loci
 - Q4: Centers and Radii Extrapolation
 - Q5: Equations from Moduli Differences
- **4. Synthesis:** The Anatomy of Complex Loci



Foundational Framework: Locus & Circles

$$|z - z_0| = r$$

The locus of z that satisfies the equation above, where z_0 is a fixed complex number and r is a fixed positive real number.

Key Conditions:

- **Center:** z_0 | **Radius:** r
- **On the Circle:** $|z - z_0| = r$
- **Interior of the Circle:** $|z - z_0| < r$
- **Exterior of the Circle:** $|z - z_0| > r$

The Cartesian Conversion

To find the Cartesian locus, always substitute:

$$z = x + iy \text{ (where } x, y \in \mathbb{R})$$

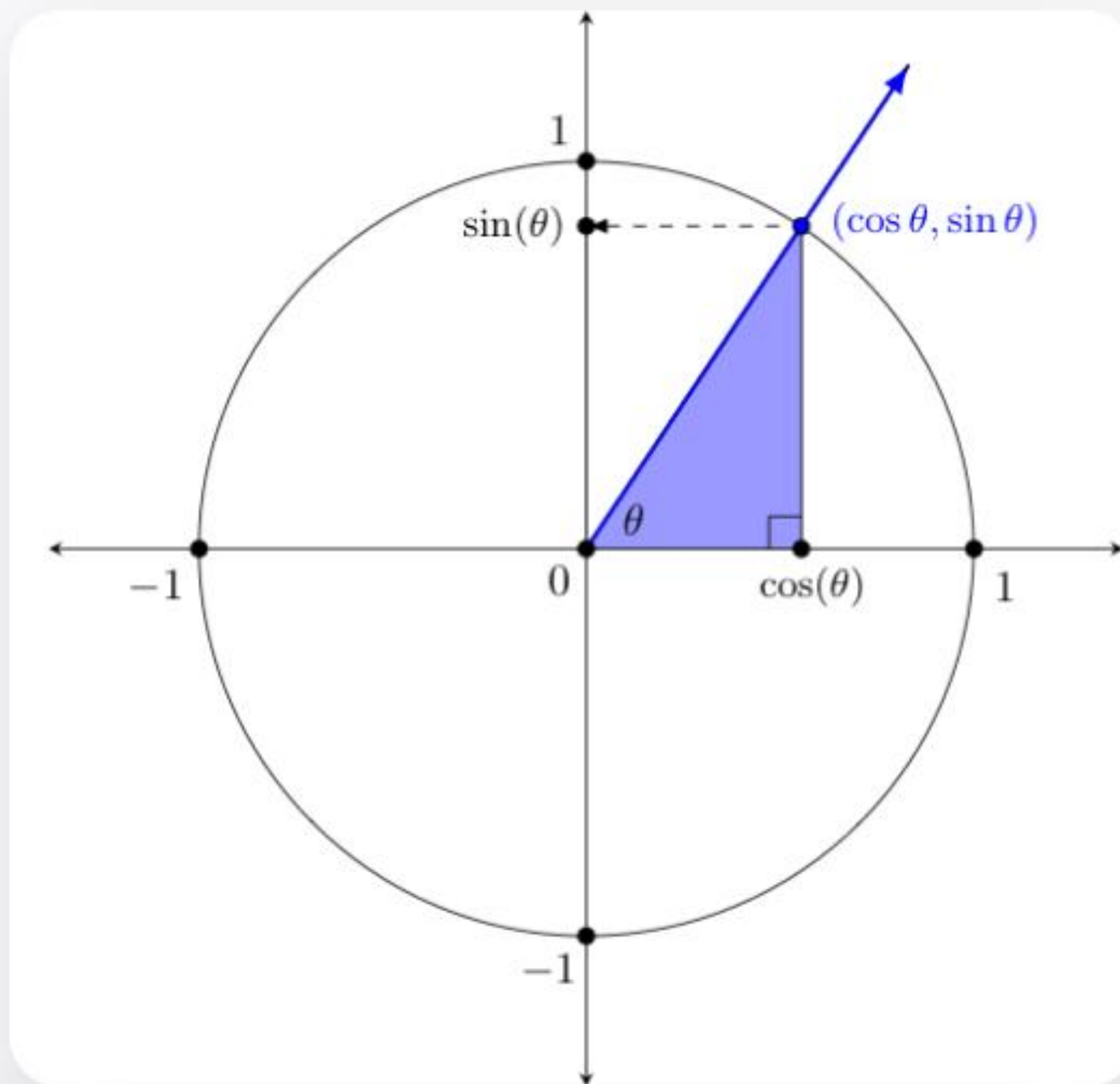
 Circle in the Argand plane with center z_0 and radius r

Example 2.18: Complex Triangles

Setup

Given: $z = 3 + 2i$ Points A, B, C represent z , iz , and $z + iz$.

Required: Represent in the Argand plane and show they form an isosceles right triangle.



Mathematical Proof

1. Find the vertices:

$$z = 3 + 2i \Rightarrow A(3, 2)$$

$$iz = i(3 + 2i) = 3i + 2i^2 = -2 + 3i \Rightarrow B(-2, 3)$$

$$z + iz = (3 + 2i) + (-2 + 3i) = 1 + 5i \Rightarrow C(1, 5)$$

2. Calculate side lengths squared ($|z_1 - z_2|^2$):

$$AB^2 = |z - iz|^2 = |(3 - (-2)) + i(2 - 3)|^2 = |5 - i|^2 = 5^2 + (-1)^2 = 26$$

$$BC^2 = |iz - (z + iz)|^2 = |-z|^2 = |-3 - 2i|^2 = (-3)^2 + (-2)^2 = 13$$

$$CA^2 = |(z + iz) - z|^2 = |iz|^2 = |-2 + 3i|^2 = (-2)^2 + 3^2 = 13$$

3. Verify geometric properties:

$$BC^2 = CA^2 = 13 \Rightarrow BC = CA \text{ (Isosceles)}$$

Example 2.19: Standardizing the Circle Equation

Setup

Given:

Locus equation

$$|3z - 5 + i| = 4.$$

Required:

Show it represents a circle; find its center and radius.

Mathematical Proof

1. Recall Standard Form:

$$|z - z_0| = r$$

2. Standardize Equation:

Factor out the coefficient of z (3):

$$|3(z - \frac{5}{3} + \frac{i}{3})| = 4$$

$$3|z - (\frac{5}{3} - \frac{i}{3})| = 4$$

$$|z - (\frac{5}{3} - \frac{i}{3})| = \frac{4}{3}$$

3. Identify Components:

Comparing with $|z - z_0| = r$:

Example 2.20: Interior of a Circle

Setup

Given: The inequality

$$|z + 2 - i| < 2.$$

Required: Show it represents interior points of a circle; find center and radius.

Mathematical Proof

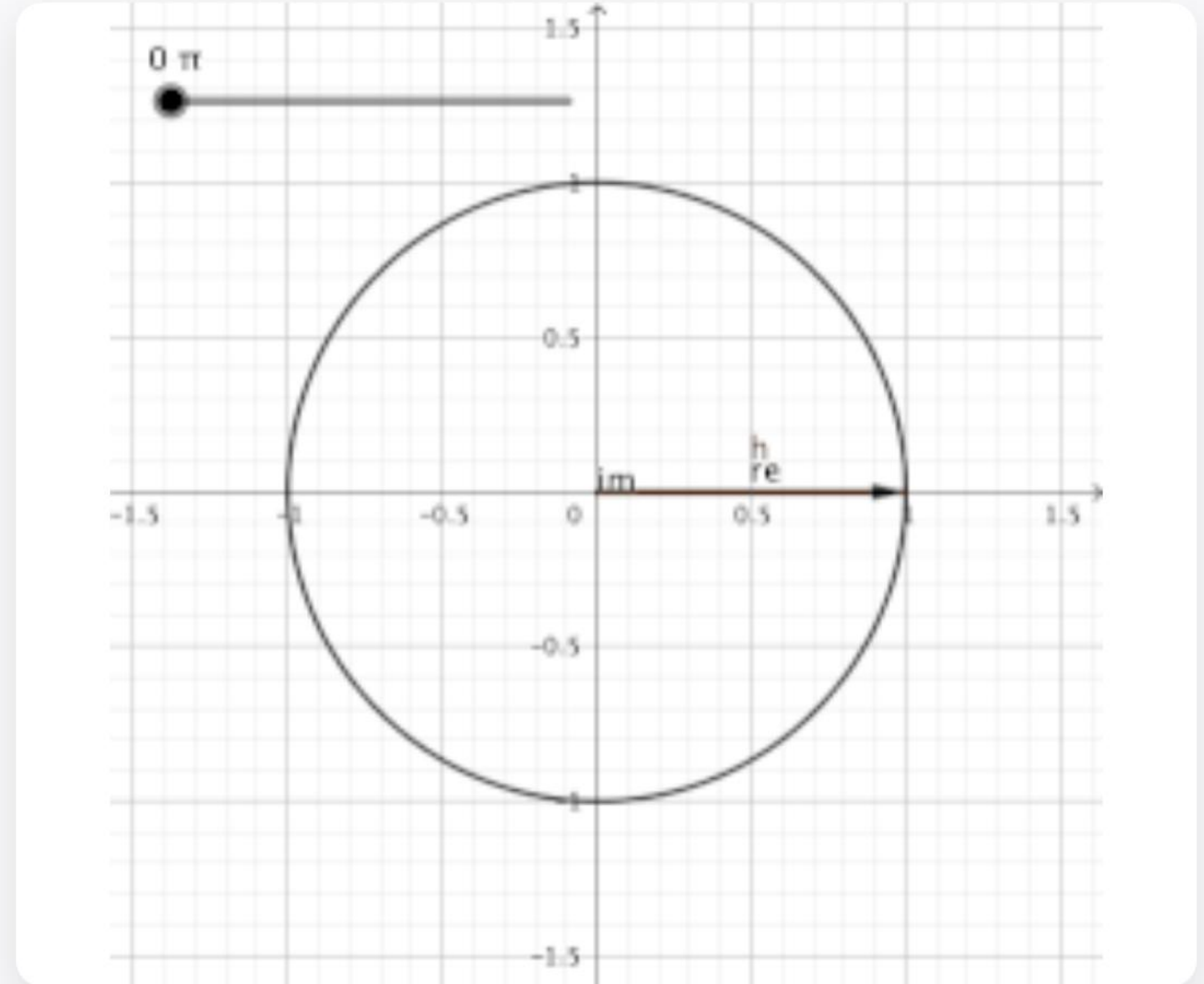
1. Recall Inequality Condition:

$|z - z_0| < r$ represents the interior of a circle.

2. Rewrite Given Inequality:

Factor out a negative sign to match $(z - z_0)$:

$$|z - (-2 + i)| < 2$$



Example 2.21: Finding the Cartesian Locus

Given: (i) $|z| = |z - i|$ (ii) $|2z - 3 - i| = 3$

Required: Obtain Cartesian form (substitute $z = x + iy$).

Problem (i)

$$|z| = |z - i|$$

$$|x + iy| = |x + i(y - 1)|$$

Square both sides:

$$x^2 + y^2 = x^2 + (y - 1)^2$$

$$x^2 + y^2 = x^2 + y^2 - 2y + 1$$

$$0 = -2y + 1 \Rightarrow 2y - 1 = 0$$

$$2y - 1 = 0$$

Problem (ii)

$$|2z - 3 - i| = 3$$

$$|2(x + iy) - 3 - i| = 3 \Rightarrow |(2x - 3) + i(2y - 1)| = 3$$

Square both sides:

$$(2x - 3)^2 + (2y - 1)^2 = 9$$

$$4x^2 - 12x + 9 + 4y^2 - 4y + 1 = 9$$

$$4x^2 + 4y^2 - 12x - 4y + 1 = 0$$

$$4x^2 + 4y^2 - 12x - 4y + 1 = 0$$

Exercise 2.6 (Q1): Locus from Modulus Ratio

Setup

Given: $z = x + iy$ and

$$\frac{z - 4i}{z + 4i} = 1.$$

Required:

Show that the locus of z is the real axis.

Mathematical Proof

1. Apply Modulus Division Property:

$$\frac{z_1}{z_2} = \frac{|z_1|}{|z_2|}$$

$$\frac{z - 4i}{z + 4i} = 1 \Rightarrow |z - 4i| = |z + 4i|$$

2. Substitute $z = x + iy$:

$$\begin{aligned} |x + iy - 4i| &= |x + iy + 4i| \\ |x + i(y - 4)| &= |x + i(y + 4)| \end{aligned}$$

3. Square & Expand:

$$\begin{aligned} x^2 + (y - 4)^2 &= x^2 + (y + 4)^2 \\ x^2 + y^2 - 8y + 16 &= x^2 + y^2 + 8y + 16 \\ -8y &= 8y \Rightarrow 16y = 0 \Rightarrow y = 0 \end{aligned}$$

Exercise 2.6 (Q2): Imaginary Condition Locus

Given: $z = x + iy$ and $\text{Im} \frac{2z+1}{iz+1} = 0$.

Required: Show the locus is $2x^2 + 2y^2 + x - 2y = 0$.

1. Substitute and Group:

$$\text{Numerator: } 2(x + iy) + 1 = (2x + 1) + i(2y)$$

Denominator:

$$i(x + iy) + 1 = ix - y + 1 = (1 - y) + ix$$

2. Rationalize Denominator:

$$\frac{(2x + 1) + i(2y)}{(1 - y) + ix} \times \frac{(1 - y) - ix}{(1 - y) - ix}$$

3. Extract ONLY the Imaginary Part:

Time-Saving Technique:

Do not calculate the real part of the numerator!

$$\begin{aligned} \text{Im} &= \frac{-x(2x + 1) + 2y(1 - y)}{(1 - y)^2 + x^2} = 0 \\ -2x^2 - x + 2y - 2y^2 &= 0 \end{aligned}$$

Multiplying by -1 yields the required locus:

$$2x^2 + 2y^2 + x - 2y = 0$$

Exercise 2.6 (Q3 i–ii): Cartesian Locus from Re/Im

Given & Required: Find the Cartesian equation for the locus of $z = x + iy$.

Part (i): $\operatorname{Re}(iz) = 3$

1. Evaluate iz :

$$iz = i(x + iy) = ix + iy^2 = -y + ix$$

2. Extract Real Part:

$$\operatorname{Re}(iz) = -y$$

3. Apply Condition:

$$-y = 3 \Rightarrow y = -3$$

$$y + 3 = 0$$

Part (ii): $\operatorname{Im}((1 - i)z + 1) = 0$

1. Evaluate Expression:

$$(1 - i)(x + iy) + 1 = (x + iy - ix - i^2 y) + 1 \\ = (x + y + 1) + i(y - x)$$

2. Extract Imaginary Part:

$$\operatorname{Im} = y - x$$

3. Apply Condition:

$$y - x = 0$$

$$x - y = 0$$

Exercise 2.6 (Q3 iii–iv): Equating Moduli & Conjugates

Given & Required: Find the Cartesian equation for the locus of $z = x + iy$.

Part (iii): $|z + i| = |z - 1|$

$$|x + i(y + 1)| = |(x - 1) + iy|$$

Square both sides:

$$x^2 + (y + 1)^2 = (x - 1)^2 + y^2$$

$$x^2 + y^2 + 2y + 1 = x^2 - 2x + 1 + y^2$$

$$2y + 1 = -2x + 1$$

$$2x + 2y = 0 \Rightarrow x + y = 0$$

$$x + y = 0$$

Part (iv): $\bar{z} = z^{-1}$

Rewrite z^{-1} as $\frac{1}{z}$:

$$\bar{z} = \frac{1}{z}$$

$$z\bar{z} = 1$$

Recall Identity: $z\bar{z} = |z|^2 = x^2 + y^2$

Substitute identity:

$$x^2 + y^2 = 1$$

$$x^2 + y^2 = 1$$

Exercise 2.6 (Q4): Extracting Centers and Radii

Required: Show equations represent circles; find center & radius. (Target Form: $|z - z_0| = r$)

Part (i): $|z - 2 - i| = 3$

Rewrite : $|z - (2 + i)| = 3$

Center: $2 + i$, Radius: 3

Part (ii): $|2z + 2 - 4i| = 2$

Factor out 2 : $2|z + 1 - 2i| = 2$
Divide by 2 : $|z - (-1 + 2i)| = 1$

Center: $-1 + 2i$, Radius: 1

Part (ii): $|2z + 2 - 4i| = 2$

Factor out 2 : $2|z + 1 - 2i| = 2$
Divide by 2 : $|z - (-1 + 2i)| = 1$

Center: $-1 + 2i$, Radius: 1

Part (iii): $|3z - 6 + 12i| = 8$

Factor out 3 : $3|z - 2 + 4i| = 8$
Divide by 3 : $|z - (2 - 4i)| = \frac{8}{3}$

Center: $2 - 4i$, Radius: $\frac{8}{3}$

Exercise 2.6 (Q5): Modulus Algebra

Given & Required: Obtain Cartesian equation for the locus of z .

Part (i): $|z - 4| = 16$

$$|(x - 4) + iy| = 16$$

Square both sides:

$$(x - 4)^2 + y^2 = 256$$

$$x^2 - 8x + 16 + y^2 = 256$$

$$x^2 + y^2 - 8x - 240 = 0$$

$$x^2 + y^2 - 8x - 240 = 0$$

Part (ii): $|z - 4|^2 - |z - 1|^2 = 16$

$$|(x - 4) + iy|^2 - |(x - 1) + iy|^2 = 16$$

$$[(x - 4)^2 + y^2] - [(x - 1)^2 + y^2] = 16$$

$$(x^2 - 8x + 16 + y^2) - (x^2 - 2x + 1 + y^2) = 16$$

$$-8x + 16 - (-2x + 1) = 16$$

$$-6x + 15 = 16 \Rightarrow -6x = 1 \Rightarrow 6x + 1 = 0$$

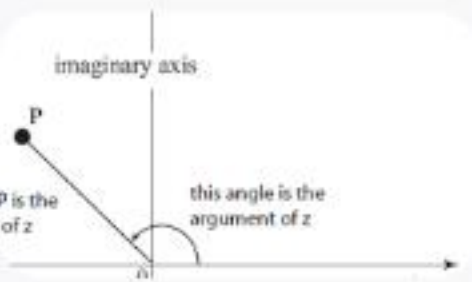
$$6x + 1 = 0$$

The Anatomy of Complex Loci: Pattern Recognition

The Ultimate Strategy: When in doubt, substitute $z = x + iy$, separate Real and Imaginary parts, and apply standard algebra.

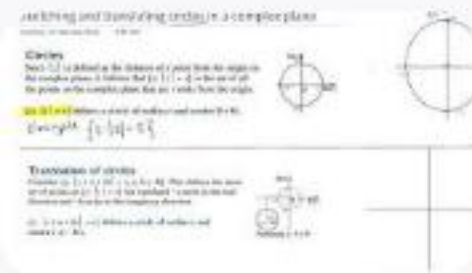
Complex Equation Form	Geometric Meaning	Algebraic Result (Cartesian)
$ z - z_0 = r$	Distance from a fixed point is constant.	Circle ($x^2 + y^2 + \dots = 0$)
$ z - z_1 = z - z_2 $	Equidistant from two fixed points.	Straight Line / Perpendicular Bisector ($ax + by + c = 0$)
$\text{Re} (f (z)) = c$ or $\text{Im} (f (z)) = c$	Real or Imaginary part meets a condition.	Various Curves (Often lines or conics depending on $f (z)$)
$z\bar{z} = c$	Square of the modulus is constant.	Circle centered at origin.

Image Sources



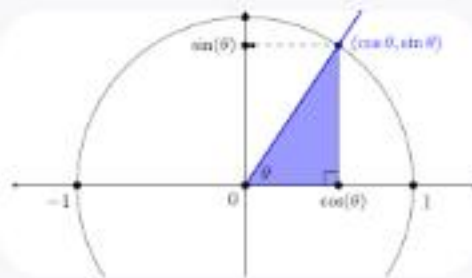
https://www.mas.ncl.ac.uk/ask/numeracy-maths-statistics/images/Modarg_diagram.png

Source: www.mas.ncl.ac.uk



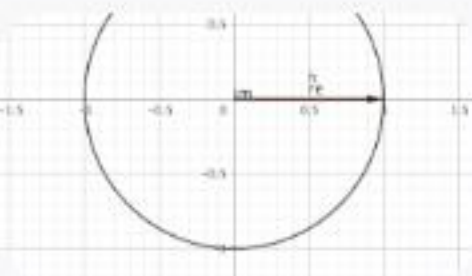
<https://i.ytimg.com/vi/kz2QZz9fvpo/sddefault.jpg>

Source: www.youtube.com



https://mathcenter.oxford.emory.edu/site/math108/trig_defs/trig_coor_defs.png

Source: mathcenter.oxford.emory.edu



<https://upload.wikimedia.org/wikipedia/commons/thumb/9/9c/Unitycircle-complex.gif/250px-Unitycircle-complex.gif>

Source: en.wikipedia.org